higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

CONCRETE STRUCTURES
NQF LEVEL 4

(12020514)

22 November 2019 (X-Paper)
09:00–12:00

This question paper consists of 5 pages.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

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QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.5) in the ANSWER BOOK.

1.1.1 When offloading reinforcing steel, the steel must be … from the delivery truck.

A dropped on a hard surface
B skidded or hoisted
C dropped on a soft surface
D skidded then dropped

1.1.2 Mild steel is … and can easily bend when it is dropped.

A light
B relatively ductile
C brittle
D a high carbon steel

1.1.3 … must be lifted with non-metallic slings.

A Mild steel rebars
B High-yield steel rebars
C Epoxy-coated rebars
D Tagged Rebars

1.1.4 … is when the corrosion cells on the surface of reinforcing steel becomes inactive when the steel is embedded in concrete.

A Acidic conditions
B Chloride conditions
C Contamination
D Passivation

1.1.5 American research showed that … improves the bond between the concrete and steel.

A a layer of salt spray
B a layer of light oil
C a layer of light rust
D a layer of mill scale

(5 × 1) (5)

1.2 Name the FOUR important things (the 'four Cs'), to prevent reinforcing steel in concrete from rusting. (4)

1.3 Explain the chemical term *hydration* when casting concrete. (1)

[10]

QUESTION 2

2.1 Briefly describe FIVE factors that must be kept in mind before tying reinforcement. (5 × 1) (5)

2.2 Illustrate by means of neat labelled sketches FIVE different types of ties that we use when tying reinforcing steel and explain the purpose and strength of each of the FIVE ties. (5 × 4) (20)

2.3 Explain step-by-step how to tie the following ties:

2.3.1 Slash tie

2.3.2 Hairpin tie

2.3.3 Splice tie

(3 × 2) (6)

2.4 Make a neat sketch of each of the following tools:

2.4.1 Tying wire nips

2.4.2 Bar wire twister

(2 × 2) (4)

[35]

QUESTION 3

Design a pre-concreting inspection checklist that will include TEN checks to be done on reinforcing. **[10]**

QUESTION 4

4.1 Give FIVE general rules that should be adhered to after concrete has been discharged from the concrete mixer up to the point when the concrete is placed. (5)

4.2 Give THREE reasons for the cause of honeycombing when placing concrete in tall walls. (3)

4.3 Give FIVE signs of possible false work and formwork failure when placing concrete on a large deck. (5)

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4.4 FIGURE 1 below shows a diagram with different slump test results that were performed for reinforced walls. The test results are numbered from (A–D).

Name and explain each of the tests outcomes (A–D) in FIGURE 1 and what solutions would you suggest to modify the mix design of the different tests if necessary.

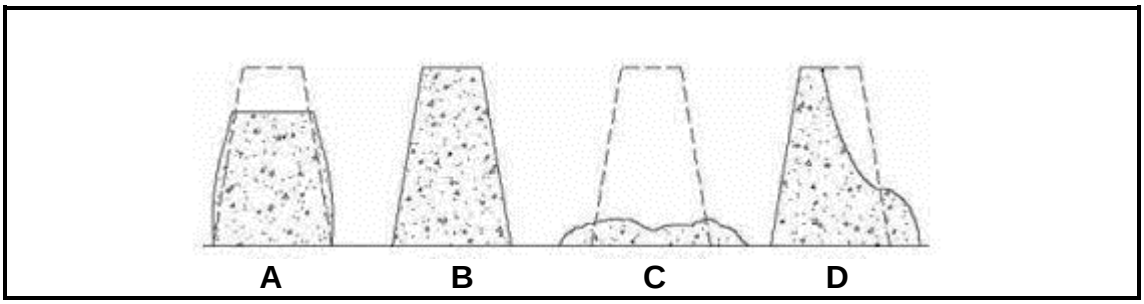


FIGURE 1 (4 × 3) (12)
[25]

QUESTION 5

5.1 The temperature readings in question number (5.1.1–5.1.3) were recorded on THREE different building sites when concrete beams were cast with a cement strength class of 42.5 R.

Interpret the readings provided in question (5.1.1–5.1.3) and categorize each of the readings in ONE of the THREE categories (Hot or normal; Cool; Cold) according to the SANS 2001- CC1 and indicate next to each category, after how many days the beam props including the cantilevers can be removed.

Write only the answers next to the question number (5.1.1–5.1.3) in the ANSWER BOOK.

5.1.1 34 ºC

5.1.2 10 ºC

5.1.3 20 ºC

(3 × 2) (6)

5.2 State FOUR factors that must be considered before removing formwork. (4)
[10]

QUESTION 6

6.1 Explain how the following formwork panels are stored especially when they are not in use:

6.1.1 Timber panels (3)

6.1.2 Steel panels (3)

6.2 Name FOUR methods to organise and store small components (4)
[10]

TOTAL: 100

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